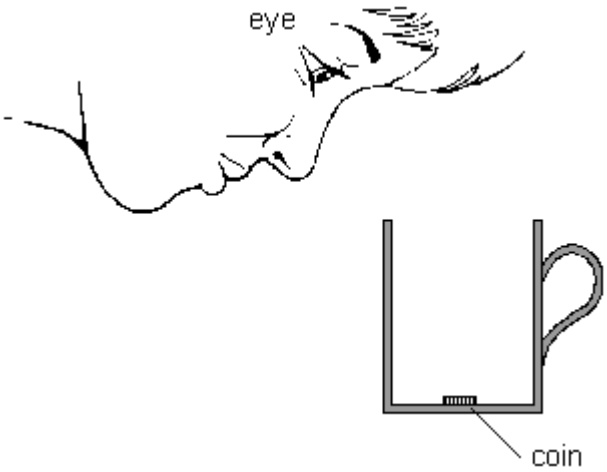
Q1.

Sophie places a coin at the bottom of an empty mug. She cannot see the coin with her eye in the position shown.



(a) Sophie fills the mug with water. Her head is in the same position as before, but now she can see part of the coin.



Draw a ray of light on the diagram to show how Sophie can see part of the coin.
Use a ruler.
Draw an arrow on the ray to show its direction.

3 marks

(b) Sophie pours some concentrated blackcurrant juice into the water.
The blackcurrant drink acts like a red filter and makes the coin look red.
Explain how a red filter works.

.....

.....

.....

2 marks
Maximum 5 marks

Q1.

(a) one mark for a ray from coin to eye, bending at the surface of the water, and not passing through the mug

both parts of the ray must slope upwards to the left
accept small discontinuities in the ray accept rays which are almost straight but which may have not been drawn with a ruler

1

one mark for accurately drawing the correct ray

*do **not** award this mark unless the first mark was also awarded the ray must touch the coin, touch the pupil of the eye, and must be continuous*
both parts of the ray must be drawn with a ruler

1

one mark for an arrow showing that the light enters the eye

1

(b) red light passes through

*answers may be in either order accept 'the filter **or** drink does not affect the red light'*
*do **not** accept 'it reflects red light'*

1

the other colours are absorbed

*accept 'only the red light passes through the filter**'***
***or '**every colour except red is absorbed' for both marks*
*do **not** accept 'the light turns red'*

1

[5]